Rajat

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SUMMARY

AI/ML Engineer and Full Stack Developer specializing in Machine Learning, NLP, Generative AI, and intelligent applications. Experienced in developing end-to-end AI solutions, LLM-powered agents, retrieval-augmented systems, predictive models, and scalable web applications using Python, FastAPI, and modern web technologies. Passionate about applying AI-driven automation to solve real-world problems and transform complex data into actionable insights.

WORK EXPERIENCE

FOKEL June 2026 – September 2026

AI Solutions Engineer Intern

- Translated business requirements into technical workflows, identifying opportunities to apply AI agents and automation to streamline business operations.
- Designed and prototyped LLM-powered solutions using prompt engineering, structured outputs, and API/tool integrations to connect AI capabilities with business systems.
- Tested and refined AI workflows for response quality, reliability, and edge cases, supporting iterative improvements in solution performance and usability.

IBM SKILLSBUILD TRAINING PROGRAM June 2025 – August 2025

AI & Machine Learning Trainee

- Completed an intensive 2-month industry training program focused on Artificial Intelligence, Machine Learning, Data Science, and Python programming.
- Gained practical experience with Python, Pandas, NumPy, Scikit-learn, Matplotlib, and Jupyter Notebook for data analysis and predictive modeling.
- Strengthened understanding of supervised machine learning algorithms, feature engineering, data visualization, and real-world AI applications through practical assignments.

PROJECTS

UniWell: Campus Health Sanctuary Preview

- Built a full-stack healthcare management platform with secure role-based portals for students, medical staff, and administrators.
- Implemented AI-powered healthcare assistance, emergency SOS broadcasting, appointment scheduling, and digital medical records workflows.
- Designed scalable authentication and secure backend architecture using Supabase with RBAC-based access control.

ResearchGPT: Multi-Agent AI Research Assistant

- Engineered a multi-agent AI system using LangGraph and Gemini 2.5 Pro, orchestrating autonomous agents for planning, information retrieval, analysis, report generation, and data visualization.
- Designed a RAG-powered knowledge pipeline using ChromaDB, document embeddings, and semantic search to generate source-grounded responses.
- Developed a scalable AI SaaS application using Next.js, Supabase, and Clerk Authentication, supporting research automation, citation-backed reporting, and interactive analytics.

Galaxy Morphology Classification

- Developed an end-to-end galaxy image classification pipeline using the Galaxy10 DECaLS dataset to classify galaxies into Elliptical, Spiral, and Irregular morphological categories.
- Benchmarked classical ML algorithms (SVM, Random Forest, XGBoost) against transfer-learning models including ResNet18, EfficientNet-B0, and MobileNetV3-Small, achieving 81.84% test accuracy with the best-performing XGBoost model.
- Implemented explainable AI using Grad-CAM for visual interpretation and developed a FastAPI prediction backend with an interactive Streamlit dashboard.

Receipts-AI: AI Agent Verification Platform

- Developed a local-first verification system for Claude Code that validates AI coding agents' completion claims against actual test execution results and the current Git repository state.
- Implemented Git fingerprint-based test verification and automated detection of test tampering, including skipped tests, weakened assertions, and deleted test cases.
- Built a modular pnpm monorepo with an Express.js and TypeScript verification server, Next.js dashboard, shared Zod schemas, CLI tooling, and Failproof policy hooks.

SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, SQL
- **AI & ML:** Machine Learning, NLP, Computer Vision, Generative AI, LLM Applications, RAG, Prompt Engineering, Scikit-Learn, Pandas, NumPy, Matplotlib, XGBoost, OpenCV
- **Frameworks & Backend:** FastAPI, Flask, LangChain, LangGraph, Next.js, Express.js, Tailwind CSS
- **Databases & Tools:** PostgreSQL, Supabase, Convex, SQLite, ChromaDB, Git, GitHub, REST APIs, Docker

- **Systems & Workflows:** Agentic Workflows, Event-Driven Architectures, Real-Time Systems, Authentication & Authorization, Full-Stack Development

EDUCATION

Guru Gobind Singh Indraprastha University 2023 – Present
Bachelor of Technology (AI&ML) – 8.2 GPA

School of Excellence, Sec-17 Rohini 2022 – 2023
Senior Secondary – 80%